

MODULE 19

Data Visualization and Storytelling

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Data Visualization and Storytelling

Visualization Principles

Effective visualization tells a clear story.

Choose the right chart:

Comparison: bar chart, grouped bars

Trend: line chart, area chart

Composition: pie chart, stacked bars

Distribution: histogram, box plot, violin

Relationship: scatter, bubble, heatmap

Geographic: choropleth maps

Design principles:

Simplicity, consistency, accuracy

Matplotlib Mastery

```
import matplotlib.pyplot as plt
import numpy as np

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Line chart
axes[0,0].plot(x, y, color='purple', linewidth=2, label='Trend')
axes[0,0].set_title('Sales Trend', fontsize=14, fontweight='bold')
axes[0,0].set_xlabel('Month')
axes[0,0].legend()

# Bar chart
axes[0,1].bar(categories, values, color=['#7C3AED', '#06B6D4', '#10B981'])
axes[0,1].set_title('Category Comparison')

plt.tight_layout()
plt.savefig('dashboard.png', dpi=150, bbox_inches='tight')
plt.show()
```

Seaborn Statistical Plots

```
import seaborn as sns, matplotlib.pyplot as plt

sns.set_theme(style='whitegrid', palette='husl')

# Distribution
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
sns.histplot(df['salary'], kde=True, ax=axes[0])
sns.boxplot(x='department', y='salary', data=df, ax=axes[1])
sns.violinplot(x='department', y='salary', data=df, ax=axes[2])
```

```
# Pair plot
sns.pairplot(df, hue='category', diag_kind='kde')

# Heatmap
corr = df.corr()
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0,
square=True, linewidths=0.5)
```

Plotly Interactive Visualizations

```
import plotly.express as px
import plotly.graph_objects as go

# Interactive scatter
fig = px.scatter(df, x='age', y='salary', color='department',
size='experience', hover_data=['name'],
title='Salary by Age and Department')
fig.show()

# Interactive line chart
fig = px.line(df_monthly, x='month', y='revenue', color='product',
title='Monthly Revenue by Product')
fig.update_layout(template='plotly_dark')
fig.show()

# Save as HTML
fig.write_html('chart.html')
```

Dashboard with Streamlit

```
import streamlit as st
import pandas as pd, plotly.express as px

st.title('Data Science Dashboard')
st.sidebar.header('Filters')

df = pd.read_csv('data.csv')
dept = st.sidebar.multiselect('Department', df['dept'].unique())
if dept:
    df = df[df['dept'].isin(dept)]

col1, col2, col3 = st.columns(3)
col1.metric('Total Employees', len(df))
col2.metric('Avg Salary', f'Rs.{df["salary"].mean():.0f}')
col3.metric('Avg Experience', f'{df["exp"].mean():.1f} yrs')

fig = px.histogram(df, x='salary', color='dept')
st.plotly_chart(fig)
st.dataframe(df.head(50))
```